

## Summary

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## Some notes on notation for finding limits:

- (a) **The first step in evaluating any limit is to find the form.** This is true, even if the question does not specifically ask for the form.
- (b) To find the form of  $\lim_{x \rightarrow a} \frac{f(x)}{g(x)}$ , take the limits of the numerator and denominator SEPARATELY. Do not just “plug in  $x = a$ ”. “Plugging in” can not occur until after the function is noticed as continuous.
- (c) When writing a form of a limit, we NEVER write  $\lim_{x \rightarrow a} \frac{f(x)}{g(x)} = \frac{0}{0}$ . It is the  $=$  that makes this a mathematically incorrect statement. Instead, write  $\lim_{x \rightarrow a} \frac{f(x)}{g(x)}$  is of the form  $\frac{0}{0}$ .
- (d) When evaluating limits, do NOT drop the  $\lim_{x \rightarrow a}$  until you have evaluated the limit. For example, the following is INCORRECT:

$$\lim_{x \rightarrow 2} \frac{2x^2 - 4x}{x - 2} = \frac{2x(x - 2)}{x - 2} = 2x = 4$$

Instead, we write:

$$\lim_{x \rightarrow 2} \frac{2x^2 - 4x}{x - 2} = \lim_{x \rightarrow 2} \frac{2x(x - 2)}{x - 2} = \lim_{x \rightarrow 2} 2x = 4$$